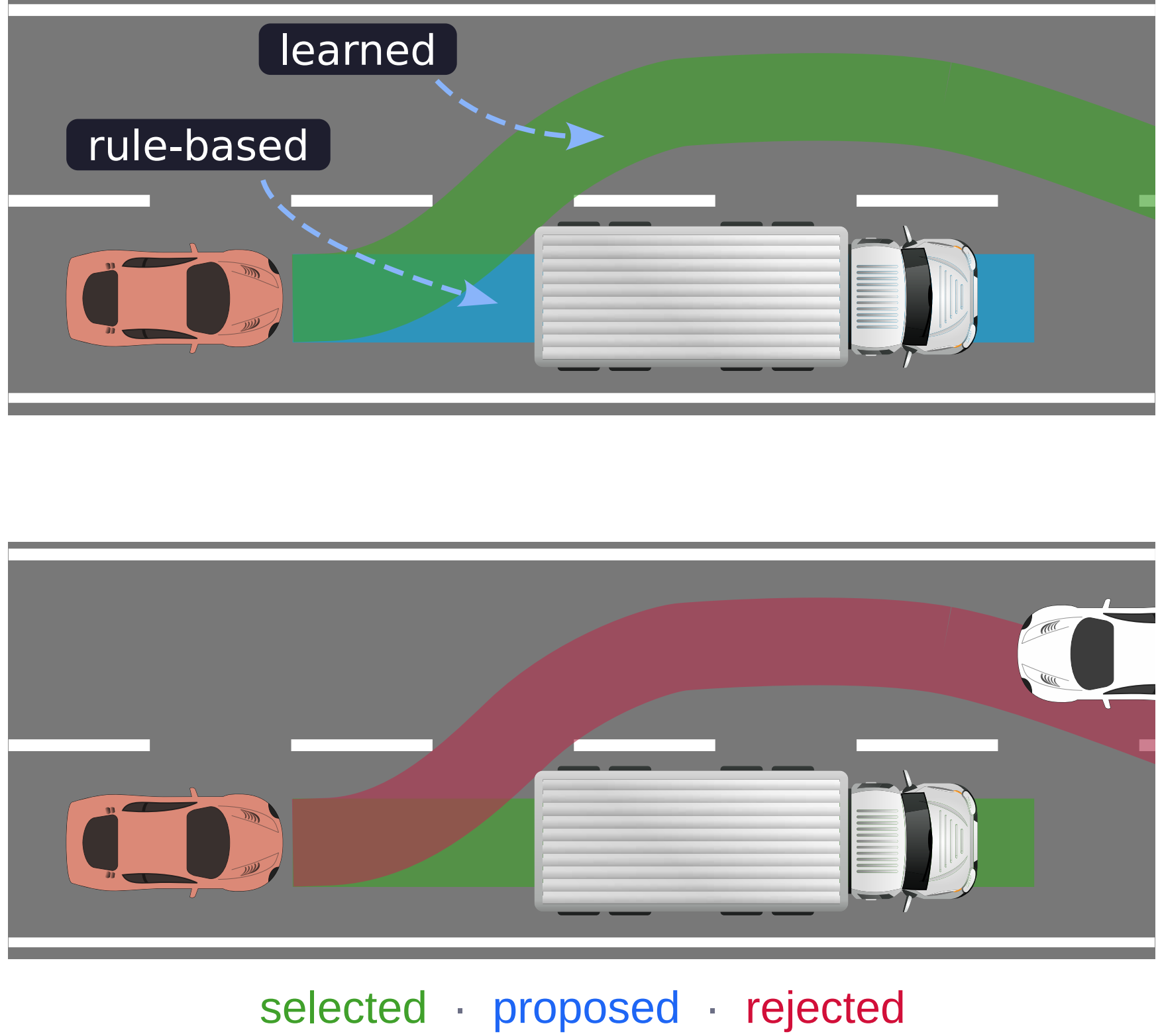


Mosaic: An Extensible Framework for Composing Rule-Based and Learned Motion Planners

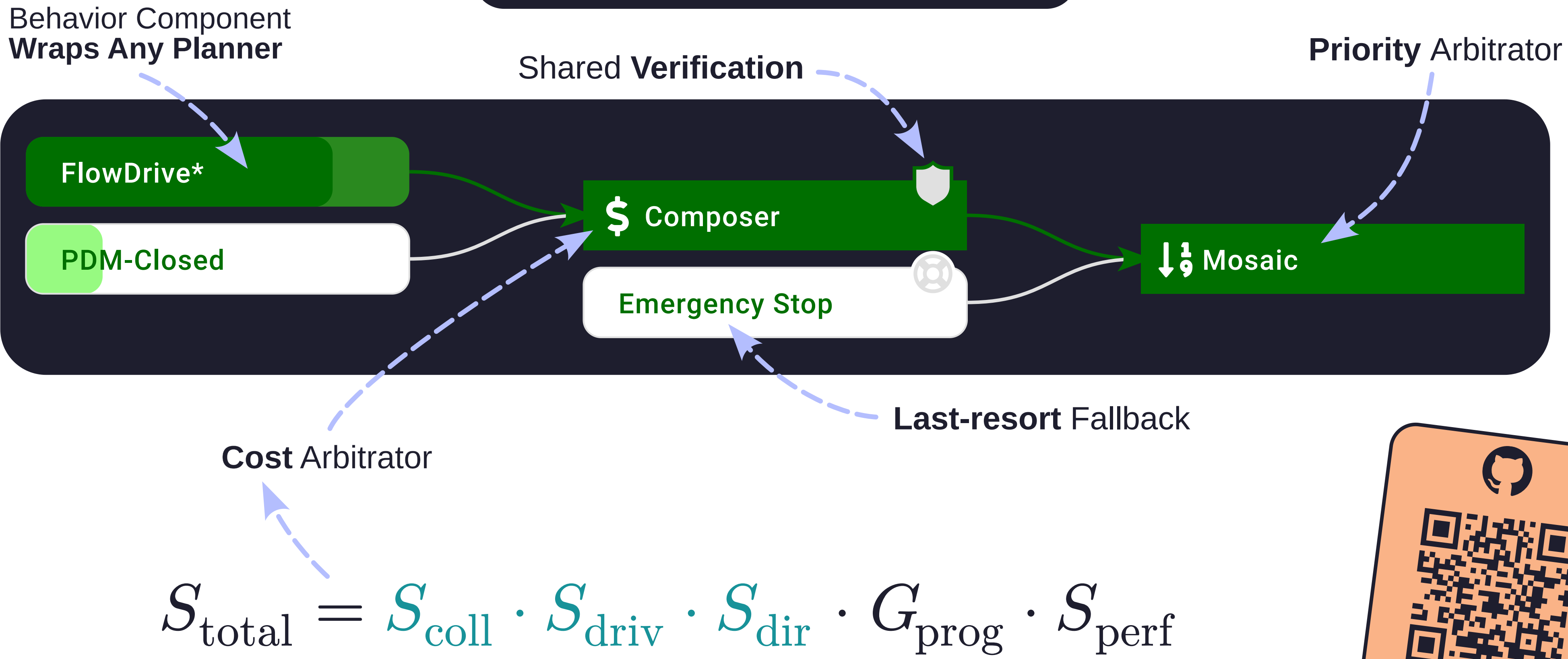
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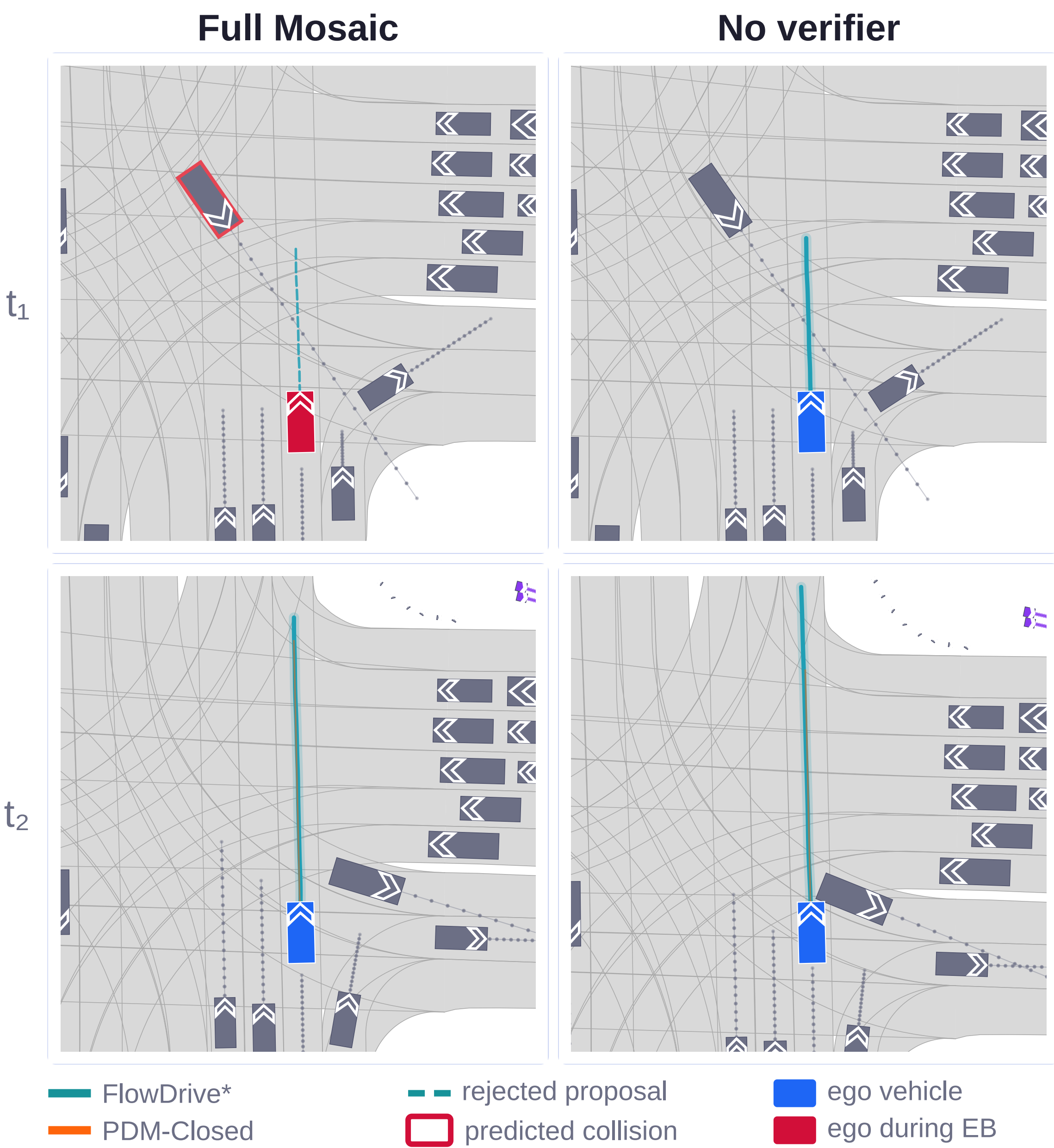
Compose & Verify



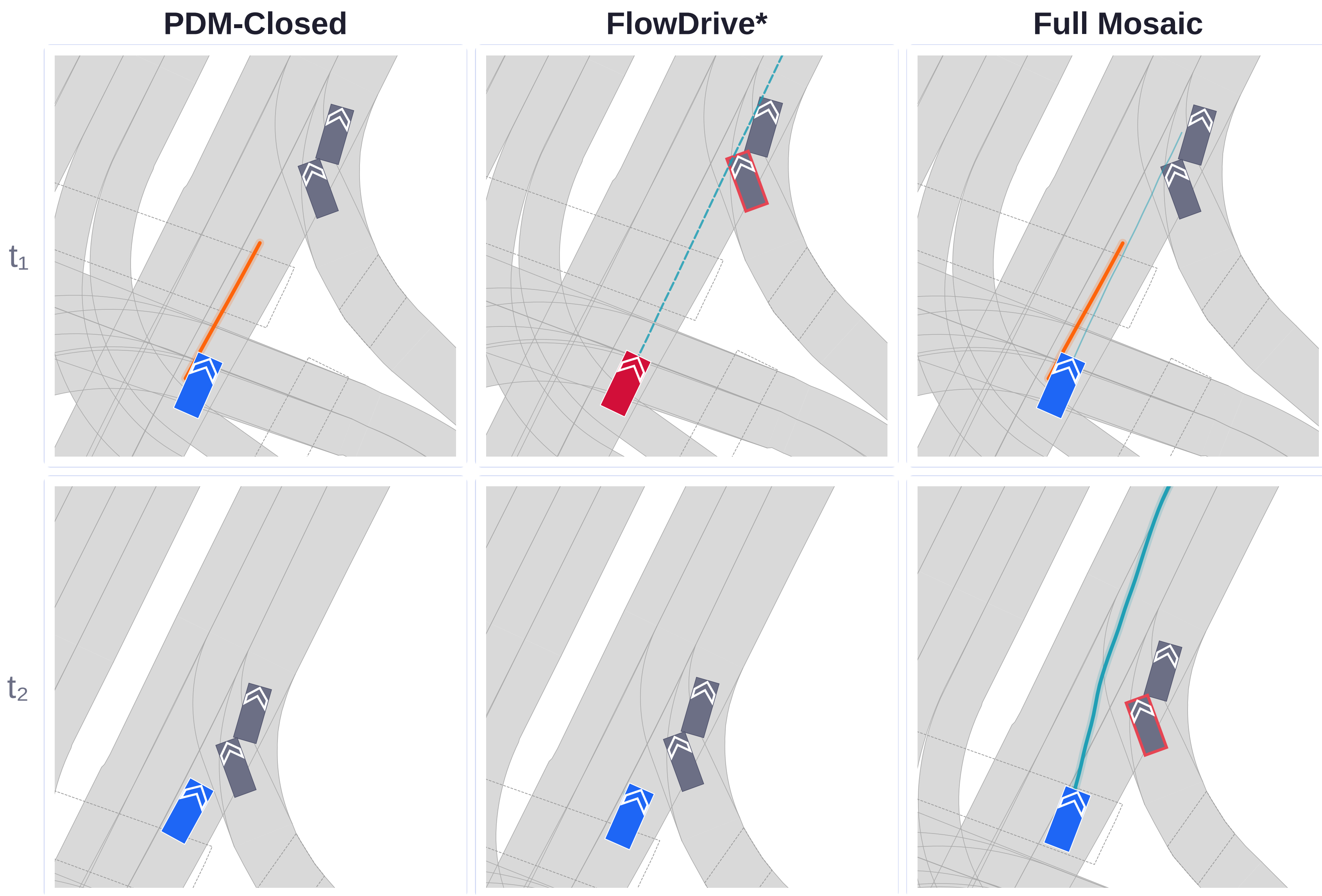
Arbitration Graph



Verification: The Safety Floor



Composition: The Performance Ceiling

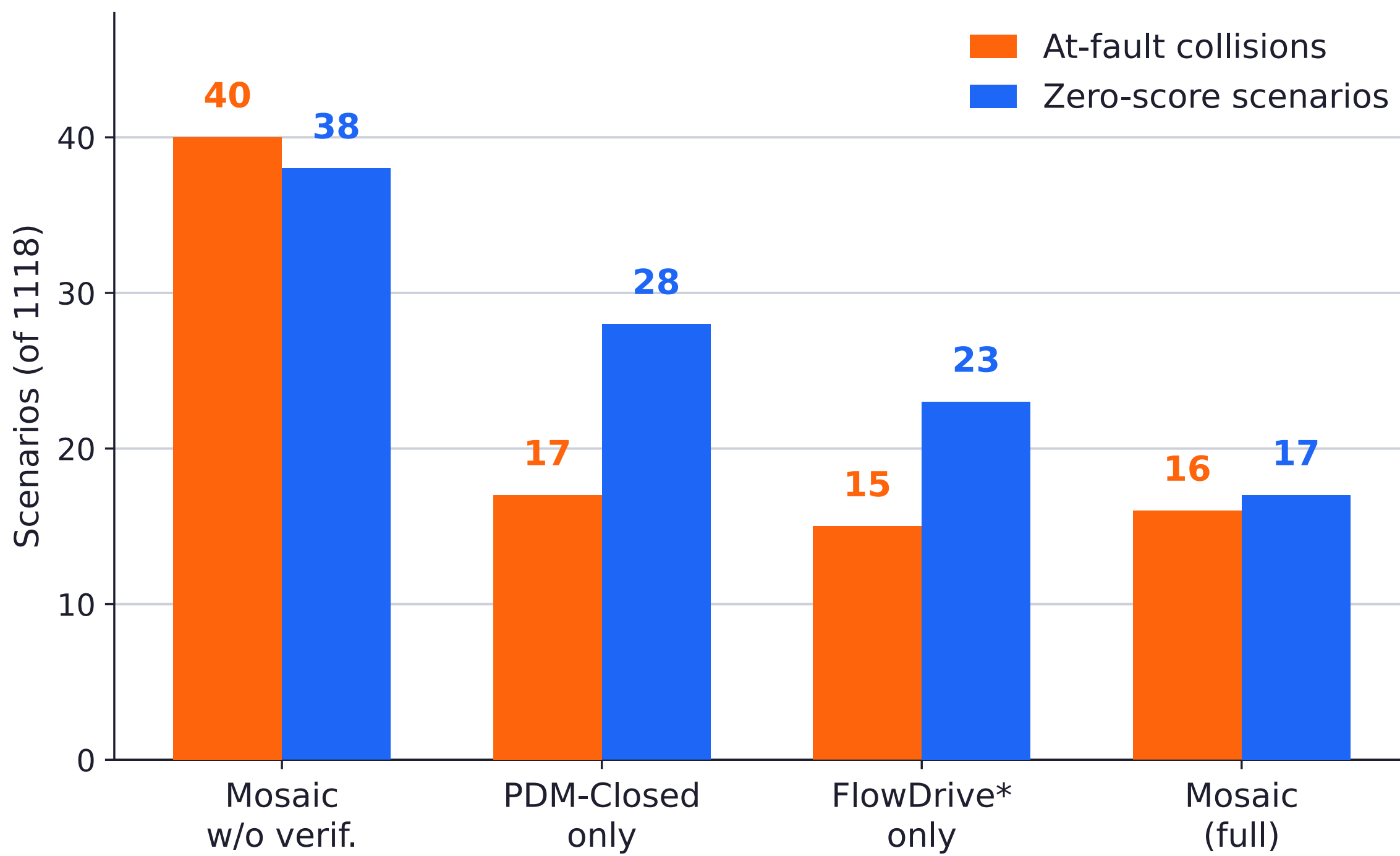


State of the Art on nuPlan

Type	Planner	Val14		interPlan ↑
		CLS-NR ↑	CLS-R ↑	
Expert	Log-replay	93.53	80.32	14.76
Learning-based	FlowDrive	91.21	85.37	36.96
	GIGAFLow†	—	93.8	—
Rule-based	PDM-Closed	92.84	92.12	41.23
Hybrid	Diffusion Planner w/ ref.	94.26	92.90	—
	FlowDrive*	94.81	92.96	44.05
Ours	Mosaic	95.56	94.18	54.10

† closed-source. Beats GIGAFLow without retraining either planner or adding data.

Two Independent Mechanisms



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